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Calculus 1

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Rock On

As of 2019 the National Geographic documentary *Free Solo* had reached 3.1 million viewers. *Free Solo* features Alex Honnold, “the only human to free solo Yosemite’s 3000 foot rock monolith, El Capitan” (Honnold). If you’re asking what does this even mean? Let me back up to help you understand the feat that Alex Honnold accomplished by free soloing El Capitan. Free soloing is the most dangerous form of climbing, it’s a style of climbing that involves climbing routes outdoors with no gear, aid, or protection whatsoever; and where falls can be fatal due to the height at which the athletes are climbing. In addition, a monolith is a large upright block of stone. Yosemite houses the largest monolith in the US, El Capitan, which is roughly twice the height of the Empire State Building. Imagine climbing El Capitan’s 3000 foot straight vertical wall with no protective gear for 3 hours and 56 minutes, while navigating small foot holds and hand holds that make your hands sweat just thinking about them (Synott). That is what Alex Honnold accomplished, and while he’s not the greatest climber in the world he is most definitely the most bold, relentless, and mentally disciplined climber of this age. The release of his documentary has widely expanded the rock climbing industry and introduced many individuals to the wonderful sport..

My interest in rock climbing began after I had stopped gymnastics in the fifth grade which was a large part of my life growing up. When I quit, I missed being able to move my body in such a dynamic, challenging way and I found my outlet through rock climbing. Throughout middle and high school, rock climbing taught me the value of belonging to a supportive community and challenged me to grow physically and mentally while connecting me to others. While brainstorming topics for this project, I didn’t immediately connect rock climbing to calculus but rock climbing is really largely an optimization equation. When a climber moves between holds, there’s so many variations of motions and paths to take to reach the finish hold. These variations in routes can be termed as beta, which is the overall strategy, sequencing, and body positioning while climbing and it is unique to every climber. The distinctive movements create a myriad of routes that are optimized for each climber. For example, a five foot climber and six foot climber would have extremely different betas and movements to get to the finish because they have to choose movements that work within their individual limits of reach, flexibility, and stability. These aspects combine to create instinctively optimized movements that minimize total work but maximize shortest distance. Shorter climbers face the challenge of reach and tend to be more dynamic in their movements, while taller climbers have the luxury of reach and can be more stable on the wall. These roles can be reversed when a climb is considered to have a “small box” where movements need to be close to the body and this is where short climbers prevail and taller climbers face the challenge of moving in limited space. These examples provide for the creation of different movements and add to the ways in which a climber forms their path to the top. Every new motion expands the variation in which a climb can be completed. In a community setting, these differences are interesting to observe as climbers interact with each other and open a space for collaboration when sharing beta and wisdom on how to finish a route. Overall, climbing is as much an individual sport as it is a team sport and it is compelling to see the ways in which different climbers optimize a specific route while considering their differences in height, strength, and flexibility. While climbing you are able to see how one discovers unconventional movements, proving the mental challenge of solving a route as well the physical challenge of completing the route in such a way that minimizes time.

In rock climbing, especially competition bouldering, athletes constantly make strategic decisions on how to move through a sequence of holds as effectively as possible. As I mentioned above, one component of climbing is time optimization and choosing the route to the top that minimizes total time and that works within the physical constraints of the athlete. Climbers can transition between fast and slow movement zones that drastically affect the overall time because different movements have different speeds such as jumping versus intricate footwork. To illustrate this concept I use a route from the Women's 2023 Prague World Cup Bouldering competition (YouTube). I examine a slab-style climb, with a relatively vertical wall, beginning with a dynamic jump (dyno) from the start to the next major volume. After the initial jump, the climber ascends diagonally toward the top using a series of holds and large volumes. The key decision lies when the climber should switch from the explosive movement of jumping to controlled upward climbing. This creates a natural optimization problem that calculus can solve. For the model, we assume the wall as two dimensional with holds representing points and two different speeds of movements; fast for dynamic and momentum driven movements and slow for controlled climbing on more technical holds. Assuming the wall as two dimensional aids in the ease of calculating the optimization problem but it is a constraint as it disregards any change in the angle of the wall or volumes which can affect trajectory. Further constraints lie in the fact that climbing is not a linear line and the calculation deals with linear descriptions of movements.

The distance between each hold is also estimated based on the known height of the wall being four meters but the exact distance is unknown. This will cause the calculation to not replicate the exact movement of a climber as they ascend the wall but it will demonstrate the overall idea of a climber needing to choose a transition point on the wall where they switch from the fast mode to the slower mode. Despite these constraints, climbers constantly make real-time optimization decisions whether or not it's consciously but modeling such decisions mathematically allows us to better understand the principles of efficient movements and route planning.

To determine the optimal transition point at which a climber should switch from dynamic movement to controlled climbing, I modeled the movement after the classic "dog running on sand and swimming in water" problem. Both problems are a time optimization problem where a subject moves faster in one medium than another.

Key Calculations

Assumptions/Variables:

- Given that the height of the wall is 4m I assumed the distance between hold one and two to be 2m and the distance between two and seven as 3m
- The rate a which a climber dynos is 1.4m/s and when they climb is 0.4m/s (OpenAI, 2025)
- Hold numbers (1-7) are modeled in the nomenclature

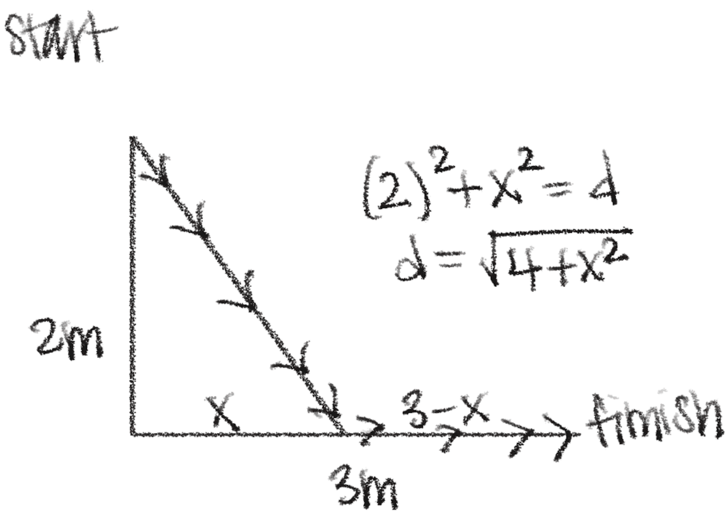
To begin, a limiting time optimization problem begins with the relationship of distance in relation to time and rate, where time is equal to distance divided by rate.

$$d = r \cdot t \rightarrow t = \frac{d}{r}$$

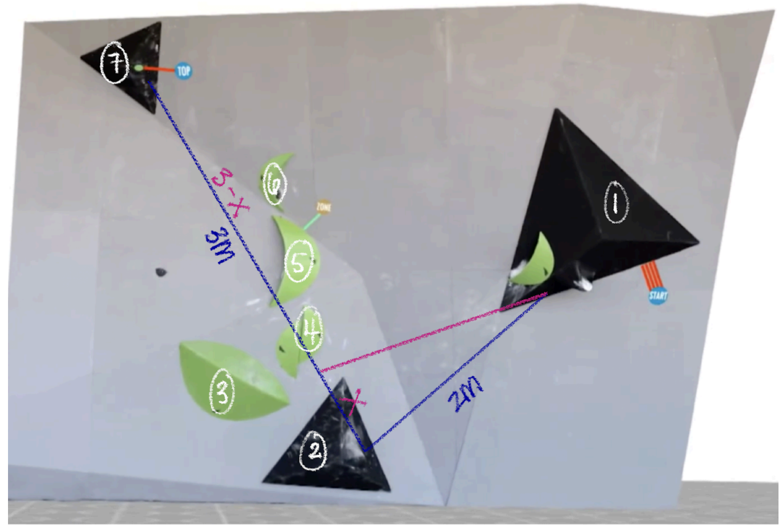
From there we can set up the equation.

$$\text{time} = \frac{\text{distance dyno}}{\text{rate dyno}} + \frac{\text{distance climbing}}{\text{rate climbing}}$$

Next, the problem is modeled separately with dimensions and unknown values.



$$(2)^2 + x^2 = 4$$
$$d = \sqrt{4+x^2}$$



Now, the values are gathered and modeled in a picture and we are able to input the values into the time function T(x) and solve for the derivative.

$$T(x) = \frac{\sqrt{4+x^2}}{1.4} + \frac{3-x}{0.4} \qquad \text{domain: } (0,3)$$

$$T(x) = \frac{7}{5}(4+x^2)^{1/2} + \frac{2}{5}(3-x) \rightarrow T(x) = \frac{7}{5}(4+x^2)^{1/2} + \frac{2}{5}(3-x)$$

$$T'(x) = \frac{7}{10}(4+x^2)^{-1/2} \cdot 2x + \frac{2}{5} \cdot -1$$

$$T'(x) = \frac{7x}{5\sqrt{4+x^2}} - \frac{2}{5} = \frac{7x - 2\sqrt{4+x^2}}{5\sqrt{4+x^2}}$$

Next, critical points are evaluated.

$T'(x)$ undefined

$$5\sqrt{4+x^2}=0$$
$$x^2\neq -4$$

$T'(x)=0$

$$7x-2\sqrt{4+x^2}=0$$
$$7x=2\sqrt{4+x^2}$$
$$49x^2=2(4+x^2)$$
$$49x^2=8+2x^2$$
$$47x^2=8$$
$$x=\sqrt{\frac{8}{47}}$$

Finally, with values using the interior minimum and boundaries of the domain, the function of time was able to be calculated and observed at which point time is minimized. The climber should transition from dynamic to controlled climbing at a point 2.587 meters away from the top hold in order to reach the top in the least amount of time.

$$3-x \qquad x=\sqrt{\frac{8}{47}}$$
$$3-\sqrt{\frac{8}{47}}=\frac{141-2\sqrt{94}}{47}\approx 2.587$$

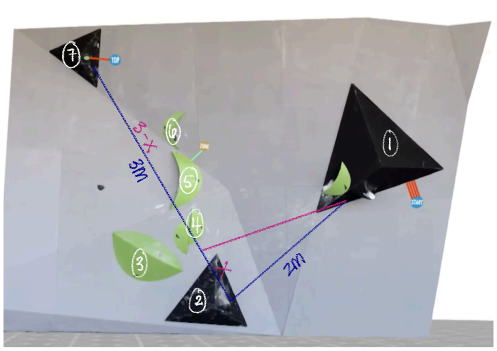
x	$T(x)$
0	8.9286
3	7.9272
$\sqrt{\frac{8}{47}}$	2.5754

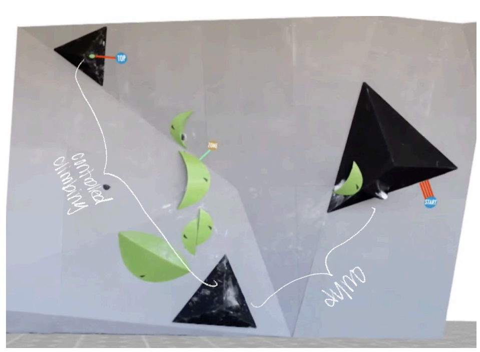
Solving this optimization problem provided a concrete solution for the climber to minimize the amount of time on the wall. The climber should transition from dynamic climbing to controlled climbing at $x = 3 - \sqrt{\frac{8}{47}}$ meters from the finish hold. The results exhibit that there is an exact point along the route where transitioning from dynamic climbing to controlled climbing minimizes total time to reach the top. In the first steps of finding a solution I created an equation related to distance and rate of the climber jumping added to the distance and rate of the climber climbing. Due to the differences in time of a climber climbing dynamically versus controlled, calculations are separated and combined into one large function to demonstrate overall time. This provided the base of the calculations and allowed me to bring out calculations further. Next, the route was modeled as a right triangle where hold two is the right angle and the start and top are the other coordinate points involved in solving for distance. The optimized point of transition will be somewhere on the line of upward controlled climbing and thus we can split the line into a section of “x” and “3 - x”. Using the Pythagorean theorem we can calculate the distance of “x” and input it into the function of time. Solving for the derivative of the function allows us to make further calculations and find the critical points at which a possible maximum or minimum could occur. Further, the domain is calculated by making the numerator equal to zero creating a domain of (0,3). From the derivative you can find the maximum and minimum, thus optimizing your equation and finding a solution. There were no undefined values on the domain but it was important to check in case there was a minimum. When the prime function was set to zero, an interior minimum point was found and was imputed back into T(x) as well as the values of the domain. Verifying domain and endpoints ensures the result is truly the absolute minimum on the given interval. Inputting the x values into T(x) provides the solution and the point at which the climber should transition from dynamic to controlled climbing. The written solution for this problem is: the climber should transition from dynamic to controlled climbing at a point 2.587 meters away from the top hold in order to reach the top in the least amount of time.. This reveals that the most efficient path is not solely dynamic climbing or controlled climbing but a calculated combination of both.

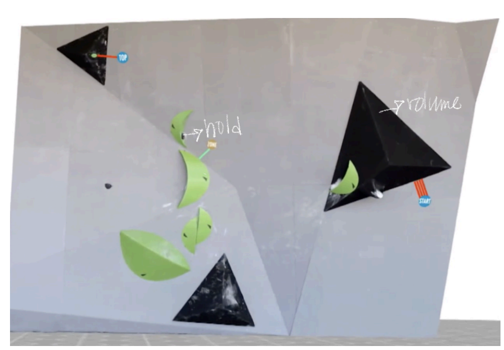
The application of calculus transformed a real world situation with competing [redacted] a precise, optimized solution. By modeling total time as a function and using derivatives to identify where the rate of change balanced out, I was able to find the exact point that minimizes the climber's travel time. It demonstrates how calculus can depict the most efficient strategy in a setting where intuition is largely used. More broadly, the problem highlights the reason for calculus being a foundation of a variety of fields as it provides the tools to analyze change, compare rates, optimize outcomes, and make decisions based on relationships between variables. Regardless of its application to the modern world and everyday life, calculus allows us to understand how small changes affect overall results making its understanding and application widely useful and relevant.

Nomenclature

- Free solo - type of free climbing that involves climbing routes with no aids or protection whatsoever
- Monolith - large upright block of stone
- Beta - overall strategy, sequencing, and body positioning a climber adopts while climbing
- Slab - climbing a surface that is less than vertical, typically leaning at an angle
- Dyno - the use of momentum and power to cross a large distance between holds, typically requiring one to jump from hold to hold and can cause body to swing upon execution
- Volume - a multisided climbing feature that attaches to the wall similar to a climbing hand hold but serves as an extension of the wall







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